

Lesson 2

Material

<https://github.com/unixfile/learning-python-2026>

Today

- ▶ Numbers and arithmetic
- ▶ Booleans and truthiness
- ▶ Reading and writing files
- ▶ Code blocks and indentation

Integers and floats

```
In : foo = 42
```

```
In : type(foo)
```

```
Out: int
```

```
In : bar = 2.71828
```

```
In : type(bar)
```

```
Out: float
```

Basic arithmetic

```
In : result = 1 + 2 - 3 * 4
```

```
In : result, type(result)
```

```
Out: (-9, int)
```

```
In : quotient = 3 / 7
```

```
In : quotient, type(quotient)
```

```
Out: (0.4285714285714285, float)
```

Power and modulo

```
In : 2 ** 5
```

```
Out: 32
```

```
In : 33 % 10
```

```
Out: 3
```

Augmented assignment

`n += 1` *# $n = n + 1$*

`q /= 3` *# $q = q / 3$*

`r %= 2` *# $r = r \% 2$*

`i++` has no meaning in Python

Booleans

```
In : to_be = True
```

```
In : not_to_be = False
```

```
In : to_be or not_to_be
```

```
Out: True
```

```
In : to_be and not_to_be
```

```
Out: False
```

Truthiness

```
In : bool(0)
```

```
Out: False
```

```
In : bool(1600)
```

```
Out: True
```

```
In : bool("")
```

```
Out: False
```

```
In : bool("Hamlet")
```

```
Out: True
```

```
In : bool(0.0)
```

```
Out: False
```

```
In : bool(0.0001)
```

```
Out: True
```

Using booleans and truthiness

```
if True:
    print("Always")
if False:
    print("Never")

if 1337:
    print("Always")
if 0:
    print("Never")
```


Reading from a file

```
f = open("foo.txt")  
content = f.read()  
f.close()  
type(content)  # str
```

Reading lines

```
f = open("bar.txt")
lines = f.readlines()
f.close()
type(lines)  # list
```

Iterating over lines

```
f = open("baz.txt")
for line in f:
    print(line)
f.close()
```

Writing to a file

```
f = open("file.txt", "w")  
f.write("Travel far, ")  
f.write("meet yourself.\n")  
f.flush()  
f.close()
```

Appending to a file

```
f = open("file.txt", "a")  
f.write("Appending...\n")  
f.flush()  
f.close()
```

Opening with scope

```
with open("file.txt", "r") as f:  
    data = f.read()
```

- ▶ File is closed automatically at end of block

Code blocks

Blocks start with : and indent 4 spaces

```
if True:
    print("banan")
    print("melon")
    for i in range(10):
        print(1 << i)
    print("kiwi")
    print("citron")
```

- Blocks can be nested